

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Importance & features of energy resources - Set 1

### Mark scheme

AQA spec 3.3.1-3.3.2; Energy resources; textbook pp. 170-182

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) <b>(1)</b></li> <li>• But not proportional: B has a high GDP but much lower energy use than A or E; C uses relatively high energy for its GDP <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                        | 2     |
| <b>01.2</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes <b>(1)</b></li> <li>• Manufacture and transport of the extra products bought uses energy (indirect use); larger houses use more energy to build <b>(1)</b></li> <li>• Affluent countries can afford energy for non-essential uses; cheap local energy (eg petrol in the USA) encourages wasteful use <b>(1)</b></li> </ul>                                                                                                                        | 3     |
| <b>01.3</b> | <i>Any 2 from the points below - 1 mark each, maximum 2.</i> <ul style="list-style-type: none"> <li>• Higher energy prices / taxes encouraging conservation <b>(1)</b></li> <li>• Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness <b>(1)</b></li> <li>• Milder climate needing less heating or air conditioning <b>(1)</b></li> <li>• Service-based economy: heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted <b>(1)</b></li> <li>• Smaller / more efficient vehicles, better public transport, better insulated buildings <b>(1)</b></li> </ul> | 2     |
| <b>01.4</b> | <i>1 mark for each point.</i> <ul style="list-style-type: none"> <li>• Decline of heavy industries such as steel and aluminium smelting in the UK <b>(1)</b></li> <li>• The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country; also efficiency improvements <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                            | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.5</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>Water: pumping, treatment / sterilisation, sewage treatment; Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                | 1     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 170-174; AQA spec 3.3.1.</p>                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <b>02.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) <b>(1)</b></li> <li>High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) <b>(1)</b></li> <li>High energy density fuels reach high temperatures needed for smelting / steam turbines <b>(1)</b></li> </ul>       | 3     |
| <b>02.2</b> | <p><i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i></p> <ul style="list-style-type: none"> <li><math>500\,000 / 28 = 17\,857\text{ kg}</math> <b>(1)</b></li> <li><math>= 17.9\text{ tonnes}</math> (accept 18; book quotes 13 t for 0.7% enriched fuel) <b>(1)</b></li> </ul>                                                                                                                                         | 2     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>The electricity they generate can power high-energy-requirement technologies <b>(1)</b></li> <li>Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage) <b>(1)</b></li> </ul>                                                                                                                                                                                  | 2     |
| <b>02.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Intermittent: not available all the time, eg tidal (but the times are predictable so alternatives can be planned) <b>(1)</b></li> <li>Unpredictable: intermittent and the timing / amount cannot be forecast accurately, eg wind, solar <b>(1)</b></li> </ul>                                                                                                                                                                                   | 2     |
| <b>02.5</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>Gaseous natural gas has a very low energy density by volume; LNG at <math>-160\text{ }^{\circ}\text{C}</math> has 160 times the energy density so a ship can carry a worthwhile amount <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                   | 1     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 175-178; AQA spec 3.3.2.</p>                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <b>03.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) <b>(1)</b></li> <li>Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably <b>(1)</b></li> <li>Wood / biofuels are renewable but depletable if forests are felled faster than they regrow <b>(1)</b></li> </ul>  | 3     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Other factors (technology, location, cost) restrict how much can be harnessed <b>(1)</b></li> <li>Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures <b>(1)</b></li> <li>New technologies or higher prices may make these usable in future <b>(1)</b></li> </ul> | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>03.3</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>• Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid <b>(1)</b></li> <li>• HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock <b>(1)</b></li> <li>• Solar: high light intensity, low cloud cover <b>(1)</b></li> <li>• Wave: coasts with strong reliable prevailing winds and a long fetch <b>(1)</b></li> <li>• Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow <b>(1)</b></li> <li>• Geothermal: hot rocks near the surface / recent volcanic activity <b>(1)</b></li> <li>• Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel <b>(1)</b></li> </ul> | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Solar PV cost fell dramatically from about 480 to 50 \$/MWh (about 90%), the steepest fall <b>(1)</b></li> <li>• Wind fell steadily from 200 to 45 <b>(1)</b></li> <li>• Gas stayed roughly constant at 60-70; solar and wind became cheaper than gas around 2015-2020 and are projected to fall further <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Development / research costs are still being paid for and there are no economies of scale in manufacture <b>(1)</b></li> <li>• They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure <b>(1)</b></li> <li>• As output grows, mass production, improved designs / efficiency and experience reduce costs <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid; running costs are very low <b>(1)</b></li> <li>• Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>04.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user <b>(1)</b></li> <li>• Some impacts (climate change) cannot yet be costed; decommissioning and waste costs are also uncertain <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>05.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s <b>(1)</b></li> <li>• Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss <b>(1)</b></li> <li>• Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                  | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise <b>(1)</b></li> <li>• Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO<sub>2</sub> / CO<sub>2</sub> / smoke <b>(1)</b></li> <li>• Pipelines causing habitat damage during construction; tanker transport risks <b>(1)</b></li> <li>• Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources <b>(1)</b></li> </ul> | 4     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) <b>(1)</b></li> <li>• Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise <b>(1)</b></li> <li>• By then depletion would reduce supply faster than replacements could be built, causing shortages <b>(1)</b></li> </ul>                                                                                                                                                                                 | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
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| 06.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Features: renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts</li> <li>• Fossil fuels: high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO<sub>2</sub>), politically concentrated (Middle East oil)</li> <li>• Nuclear: very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition</li> <li>• Renewables: low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs</li> <li>• Sustainability concepts: impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion)</li> <li>• Economic and political factors: subsidies, guaranteed prices, planning, external costs not included in price, energy security</li> <li>• Judgement: a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable; transition must start before depletion</li> </ul> | 9     |
|          | <b>Model answer - Level 3 (8 marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
|          | Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and CO <sub>2</sub> , and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|             | Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen. |          |
|             | No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it.                                                                                                                                                                                                                                                                                                                                                                                  |          |
|             | <b>Weaker answer - Level 1 (2 marks)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|             | Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy.                                                                                                                                                                                                                 |          |
|             | <b>What makes the difference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
|             | <ul style="list-style-type: none"> <li>• Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|             | <ul style="list-style-type: none"> <li>• Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|             | <ul style="list-style-type: none"> <li>• The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>06.2</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 | <b>1</b> |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |